

Notes on Varieties of Semiring Solvers

Amaury Mazoyer

May 12, 2026

CONTENTS

1	Preliminaries	2
1.1	Algebraic Structures	2
1.2	Categories	4
1.3	Monads	8
1.4	2-categories	13
1.5	Multicategories	14
1.6	Free algebras	16
2	Some free algebras	18
2.1	Semigroups	18
2.2	A first combination	22
2.3	Monoids	28
A	Daily recap	31
B	TODO	31
C	Questions for Ohad	32

Definition 3 (Group)

A *group* is a set G equipped with an associative binary operation $\cdot : G \times G \rightarrow G$, an identity element $e \in G$, and an inverse operation such that for all $a, b, c \in G$:

- $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ (associativity)
- $e \cdot a = a \cdot e = a$ (identity element)
- For each $a \in G$, there exists an element $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$ (inverse element)

A group is called *abelian* if its operation is commutative, i.e., for all $a, b \in G$, $a \cdot b = b \cdot a$.

Definition 4 (Semiring)

A *semiring* is a set R equipped with two binary operations: addition $+$ and multiplication $\cdot$, such that:

- $(R, +)$ is a commutative monoid with identity element 0.
- $(R, \cdot)$ is a monoid with identity element 1.
- Multiplication distributes over addition from both sides: for all $a, b, c \in R$,

$$\begin{aligned} a \cdot (b + c) &= a \cdot b + a \cdot c, \\ (a + b) \cdot c &= a \cdot c + b \cdot c. \end{aligned}$$

- The additive identity 0 is an absorbing element for multiplication: for all $a \in R$, $0 \cdot a = a \cdot 0 = 0$.

Definition 5 (Operad)

An *operad* $\mathcal{O}$ consists of:

- A collection of sets $\{\mathcal{O}_n\}_{n \geq 0}$, where $\mathcal{O}_n$ is the set of n -ary operations,
- A composition function $\gamma : \mathcal{O}_n \times \mathcal{O}_{k_1} \times \cdots \times \mathcal{O}_{k_n} \rightarrow \mathcal{O}_{k_1 + \cdots + k_n}$,
- An identity element $e \in \mathcal{O}_1$,

satisfying the following axioms:

- Associativity: For all $f \in \mathcal{O}_n$, $g_i \in \mathcal{O}_{k_i}$, and $h_{ij} \in \mathcal{O}_{m_{ij}}$, we have

$$\gamma(f; g_1, \dots, g_n) = f(g_1(h_{11}, \dots, h_{1k_1}), \dots, g_n(h_{n1}, \dots, h_{nk_n})).$$

- Identity: For all $f \in \mathcal{O}_n$, we have

$$f(e, \dots, e) = f.$$

Definition 6 (Algebraic Theory)

An *algebraic theory* $\mathcal{T}$ consists of:

- A collection of sets $\{\mathcal{T}_n\}_{n \geq 0}$, where $\mathcal{T}_n$ is the set of n -ary operations,
- A collection of equations between terms built from these operations,

such that the operations and equations define a class of algebras (models) that satisfy the given equations.

Example (Operad from an Algebraic Theory)

Given an algebraic theory $\mathcal{T}$, we can construct an operad $\mathcal{O}^{\mathcal{T}}$ where:

- $\mathcal{O}_n^{\mathcal{T}}$ consists of the n -ary operations in $\mathcal{T}$,
- The composition function γ is defined by substituting operations according to the equations of the theory,
- The identity element e corresponds to the identity operation in the theory.

1.2 Categories**Definition 7 (Category)**

A *category* $\mathcal{C}$ consists of:

- A class of objects $\text{Ob}(\mathcal{C})$,
- For each pair of objects $a, b \in \text{Ob}(\mathcal{C})$, a set of morphisms $\text{Hom}_{\mathcal{C}}(a, b)$,
- For each object $a \in \text{Ob}(\mathcal{C})$, an identity morphism $\text{id}_a \in \text{Hom}_{\mathcal{C}}(a, a)$,
- A composition operation $\circ$ such that for morphisms $f : a \rightarrow b$ and $g : b \rightarrow c$, there is a morphism $g \circ f : a \rightarrow c$,

satisfying the following axioms:

- Associativity: For morphisms $f : a \rightarrow b$, $g : b \rightarrow c$, and $h : c \rightarrow d$, we have $h \circ (g \circ f) = (h \circ g) \circ f$.
- Identity: For any morphism $f : a \rightarrow b$, we have $\text{id}_b \circ f = f = f \circ \text{id}_a$.

Example (Set Category)

The category **Set** has sets as objects and functions as morphisms. The identity morphism for a set a is the identity function $\text{id}_a : a \rightarrow a$ defined by $\text{id}_a(x) = x$ for all $x \in a$. Composition of morphisms is given by function composition.

Definition 8 (Product)

Given two objects a and b in a category $\mathcal{C}$, a *product* of a and b is an object $a \times b$ together with two morphisms $\pi_a : a \times b \rightarrow a$ and $\pi_b : a \times b \rightarrow b$ such that for any object c and morphisms $f : c \rightarrow a$ and $g : c \rightarrow b$, there exists a unique morphism $\langle f, g \rangle : c \rightarrow a \times b$ making the following diagram commute:

$$\begin{array}{ccccc}
 & & c & & \\
 & f \swarrow & \downarrow \langle f, g \rangle & \searrow g & \\
 a & \xleftarrow{\pi_a} & a \times b & \xrightarrow{\pi_b} & b
 \end{array}$$

Definition 9 (Coproduct)

Given two objects a and b in a category $\mathcal{C}$, a *coproduct* of a and b is an object $a + b$ together with two morphisms $\iota_a : a \rightarrow a + b$ and $\iota_b : b \rightarrow a + b$ such that for any object c and morphisms $f : a \rightarrow c$ and $g : b \rightarrow c$, there exists a unique morphism $[f, g] : a + b \rightarrow c$ making the following diagram commute:

$$\begin{array}{ccccc} a & \xrightarrow{\iota_a} & a + b & \xleftarrow{\iota_b} & b \\ & \searrow f & \downarrow [f, g] & \swarrow g & \\ & & c & & \end{array}$$

Definition 10 (Monoidal Category)

A *monoidal category* is a category $\mathcal{C}$ equipped with a bifunctor $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$, an object I called the unit object, and natural isomorphisms (called associators and unitors) that satisfy certain coherence conditions. Specifically, for all objects a, b, c in $\mathcal{C}$, there are natural isomorphisms:

- Associator: $\alpha_{a,b,c} : (a \otimes b) \otimes c \rightarrow a \otimes (b \otimes c)$,
- Left unitor: $\lambda_a : I \otimes a \rightarrow a$,
- Right unitor: $\rho_a : a \otimes I \rightarrow a$,

such that the following diagrams commute:

$$\begin{array}{ccc} ((a \otimes b) \otimes c) \otimes d \xrightarrow{\alpha_{a,b,c} \otimes 1_d} (a \otimes (b \otimes c)) \otimes d \xrightarrow{\alpha_{a,b \otimes c,d}} a \otimes ((b \otimes c) \otimes d) \\ \alpha_{a,b,c \otimes d} \uparrow & & 1_a \otimes \alpha_{b,c,d} \uparrow \\ (a \otimes b) \otimes (c \otimes d) & & a \otimes (b \otimes (c \otimes d)) \end{array}$$

$$(I \otimes a) \otimes b \xrightarrow{\alpha_{I,a,b}} I \otimes (a \otimes b) \xleftarrow{\rho_a} a = a \otimes I \xrightarrow{\lambda_a} a \otimes I \xrightarrow{\alpha_{a,I,I}} (a \otimes I) \otimes I$$

Definition 11 (Functor)

Given two categories $\mathcal{C}$ and $\mathcal{D}$, a *functor* $F : \mathcal{C} \rightarrow \mathcal{D}$ consists of:

- A mapping $F : \text{Ob}(\mathcal{C}) \rightarrow \text{Ob}(\mathcal{D})$ that assigns to each object a in $\mathcal{C}$ an object $F(a)$ in $\mathcal{D}$,
- For each pair of objects $a, b \in \text{Ob}(\mathcal{C})$, a mapping $F : \text{Hom}_{\mathcal{C}}(a, b) \rightarrow \text{Hom}_{\mathcal{D}}(F(a), F(b))$ that assigns to each morphism $f : a \rightarrow b$ in $\mathcal{C}$ a morphism $F(f) : F(a) \rightarrow F(b)$ in $\mathcal{D}$,

such that for all objects a, b, c in $\mathcal{C}$ and morphisms $f : a \rightarrow b$, $g : b \rightarrow c$, the following conditions hold:

- $F(g \circ f) = F(g) \circ F(f)$ (preservation of composition),
- $F(\text{id}_a) = \text{id}_{F(a)}$ (preservation of identity).

For a category $\mathcal{C}$, the *identity functor* $\text{Id}_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C}$ is defined by $\text{Id}_{\mathcal{C}}(a) = a$ for all objects a and $\text{Id}_{\mathcal{C}}(f) = f$ for all morphisms f .

- A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is *faithful* if, for every pair of objects $a, b \in \text{Ob}(\mathcal{C})$, the map $F : \text{Hom}_{\mathcal{C}}(a, b) \rightarrow \text{Hom}_{\mathcal{D}}(F(a), F(b))$ is injective.
- A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is *full* if, for every pair of objects $a, b \in \text{Ob}(\mathcal{C})$, the map $F : \text{Hom}_{\mathcal{C}}(a, b) \rightarrow \text{Hom}_{\mathcal{D}}(F(a), F(b))$ is surjective.
- A functor that is both full and faithful is called *fully faithful*.

Notation: We often denote the image of an object a under a functor F as Fa instead of $F(a)$ and the image of a morphism $f : a \rightarrow b$ as $Ff : Fa \rightarrow Fb$.

Example (Diagonal functor)

Given a category $\mathcal{C}$, the *diagonal functor* $\Delta : \mathcal{C} \rightarrow \mathcal{C} \times \mathcal{C}$ is defined by $\Delta(a) = (a, a)$ for each object a in $\mathcal{C}$ and $\Delta(f) = (f, f)$ for each morphism $f : a \rightarrow b$ in $\mathcal{C}$.

Definition 12 (Natural Transformation)

Given two functors $F, G : \mathcal{C} \rightarrow \mathcal{D}$ between categories $\mathcal{C}$ and $\mathcal{D}$, a *natural transformation* $\eta : F \rightarrow G$ consists of a family of morphisms $\{\eta_a : Fa \rightarrow Ga\}_{a \in \text{Ob}(\mathcal{C})}$ such that for every morphism $f : a \rightarrow b$ in $\mathcal{C}$, the following diagram commutes:

$$\begin{array}{ccc} Fa & \xrightarrow{Ff} & Fb \\ \eta_a \downarrow & & \downarrow \eta_b \\ Ga & \xrightarrow{Gf} & Gb \end{array}$$

Definition 13 (Adjunction)

Given two categories $\mathcal{C}$ and $\mathcal{D}$, a functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is a *left adjoint* to a functor $G : \mathcal{D} \rightarrow \mathcal{C}$ (and G is a *right adjoint* to F) if there is a natural isomorphism:

$$\text{Hom}_{\mathcal{D}}(Fa, b) \cong \text{Hom}_{\mathcal{C}}(a, Gb)$$

for all objects a in $\mathcal{C}$ and b in $\mathcal{D}$. This means that for every morphism $f : Fa \rightarrow b$ in $\mathcal{D}$, there is a unique morphism $g : a \rightarrow Gb$ in $\mathcal{C}$ such that the following diagram commutes:

$$\begin{array}{ccc} Fa & \xrightarrow{f} & b \\ \uparrow F & & \downarrow G \\ a & \xrightarrow{g} & Gb \end{array}$$

We often denote this adjunction by $F \dashv G$.

There are actually several different but equivalent ways to define adjunctions. They are linked thanks to the Yoneda lemma.

One important definition is that of the universal arrow from MacLane's *Categories for the Working Mathematician* [Mac71].

Example (Adjunction from Universal Arrows)

Given a functor $U : A \rightarrow C$ such that for every object $c \in C$, there exists a universal arrow from c to U , we can construct an adjunction $F \dashv U$ where:

- The functor $F : C \rightarrow A$ is defined by sending each object c in C to the object Fc in A that appears in the universal arrow from c to U , and sending each morphism $f : c \rightarrow c'$ in C to the unique morphism $Ff : Fc \rightarrow Fc'$ in A that makes the corresponding diagram commute.
- The unit $\eta : \text{Id}_C \rightarrow UF$ is given by the family of morphisms $\{\eta_c : c \rightarrow UFc\}_{c \in C}$ that are part of the universal arrows.

Universal arrows have more uses than just defining adjunctions. Here are other examples.

Example (Defining the coproduct with an universal arrow)

Let A be a category, and let $a, b \in A$. A *coproduct* of a and b is a universal arrow from (a, b) to the diagonal functor $\Delta : A \rightarrow A \times A$, where $\Delta(c) = (c, c)$ for any object $c \in A$.

$$\begin{array}{ccc} a & \xrightarrow{\iota_a} & a + b & \xleftarrow{\iota_b} & b \\ & \searrow f & \downarrow [f, g] & \swarrow g & \\ & & c & & \end{array} \quad \sim \quad \begin{array}{ccc} (a, b) & \xrightarrow{\eta_{(a,b)}} & \Delta(a + b) \\ & \searrow (f, g) & \downarrow \Delta[f, g] \\ & & \Delta c & \end{array}$$

We have $\eta_{(a,b)} = (\iota_a, \iota_b)$.

Example (Defining the product with an universal arrow)

Let A be a category, and let $a, b \in A$. A *product* of a and b is a universal arrow from the diagonal functor $\Delta : A \rightarrow A \times A$ to (a, b) .

$$\begin{array}{ccc} & c & \\ f \swarrow & \downarrow \langle f, g \rangle & \searrow g \\ a & \xleftarrow{\pi_a} & a \times b & \xrightarrow{\pi_b} & b \end{array} \quad \sim \quad \begin{array}{ccc} \Delta c & \xrightarrow{\varepsilon_{(a,b)}} & \Delta(a \times b) \\ & \searrow (f, g) & \downarrow (\pi_a, \pi_b) \\ & & (a, b) \end{array}$$

We have $\varepsilon_{(a,b)} = \langle f, g \rangle$.

1.3 Monads

Definition 15 (Monad)

A *monad* on a category $\mathcal{C}$ is a triple (T, η, μ) where:

- $T : \mathcal{C} \rightarrow \mathcal{C}$ is a functor,
- $\eta : \text{Id}_{\mathcal{C}} \rightarrow T$ is a natural transformation called the unit,
- $\mu : T^2 \rightarrow T$ is a natural transformation called the multiplication,

such that the following diagrams commute:

$$\begin{array}{ccc} T & \xrightarrow{T\eta} & T^2 \\ \eta T \downarrow & \searrow & \downarrow \mu \\ T^2 & \xrightarrow{\mu} & T \end{array} \quad \begin{array}{ccc} T^3 & \xrightarrow{T\mu} & T^2 \\ \mu T \downarrow & & \downarrow \mu \\ T^2 & \xrightarrow{\mu} & T \end{array}$$

Example (Monad from an Adjunction)

Given an adjunction $F \dashv G$ between categories $\mathcal{C}$ and $\mathcal{D}$, we can construct a monad on $\mathcal{C}$ as follows:

- The functor $T : \mathcal{C} \rightarrow \mathcal{C}$ is defined by $T = GF$.
- The unit $\eta : \text{Id}_{\mathcal{C}} \rightarrow T$ is given by the natural transformation that assigns to each object a in $\mathcal{C}$ the morphism $\eta_a : a \rightarrow GFa$ corresponding to the identity morphism on Fa under the adjunction.
- The multiplication $\mu : T^2 \rightarrow T$ is given by the natural transformation that assigns to each object a in $\mathcal{C}$ the morphism $\mu_a : GF GFa \rightarrow GFa$ corresponding to the composition of morphisms in $\mathcal{D}$ under the adjunction.

Definition 16 (Distributive law between monads)

Given two monads (T, η^T, μ^T) and (S, η^S, μ^S) on a category $\mathcal{C}$, a *distributive law* of T over S is a natural transformation $\lambda : TS \rightarrow ST$ such that the following diagrams commute:

$$\begin{array}{ccc} & T & \\ T\eta^S \swarrow & & \searrow \eta^{ST} \\ TS & \xrightarrow{\lambda} & ST \end{array} \quad \begin{array}{ccc} & S & \\ \eta^TS \swarrow & & \searrow S\eta^T \\ TS & \xrightarrow{\lambda} & ST \end{array}$$

$$\begin{array}{ccccc} T^2S & \xrightarrow{T\lambda} & TST & \xrightarrow{\lambda T} & ST^2 \\ \mu^TS \downarrow & & & & \downarrow S\mu^T \\ TS & \xrightarrow{\lambda} & ST & & \end{array} \quad \begin{array}{ccccc} TS^2 & \xrightarrow{\lambda S} & STS & \xrightarrow{S\lambda} & S^2T \\ T\mu^S \downarrow & & & & \downarrow \mu^{ST} \\ TS & \xrightarrow{\lambda} & ST & & \end{array}$$

This law induces a composite monad ST on $\mathcal{C}$ with

- unit $\eta^{ST} = \eta^S \circ \eta^T$
- multiplication $STST \xrightarrow{S\lambda T} SSTT \xrightarrow{\mu^S \mu^T} ST$

Definition 17 (Algebra over a monad)

Let (T, η, μ) be a monad on a category $\mathcal{C}$.

An *algebra over the monad T* (or simply a T -algebra) consists of:

- An object A in $\mathcal{C}$,
- A morphism $a : TA \rightarrow A$ in $\mathcal{C}$, called the structure map, such that the following diagrams commute:

$$\begin{array}{ccc} A & \xrightarrow{\eta_A} & TA \\ & \searrow \text{id}_A & \downarrow a \\ & & A \end{array} \quad \begin{array}{ccc} T^2A & \xrightarrow{Ta} & TA \\ \mu_A \downarrow & & \downarrow a \\ TA & \xrightarrow{a} & A \end{array}$$

The left one is called the unit property, and the right one is called the action property.

We'll now reformulate the definition of monads in terms of Kleisli categories and extension operators.

Definition 18 (Kleisli category)

Given a monad (T, η, μ) on a category $\mathcal{C}$, the *Kleisli category* $\mathcal{C}_T$ has:

- The same objects as $\mathcal{C}$,
- For each pair of objects a, b , the morphisms from a to b in $\mathcal{C}_T$ are given by $\text{Hom}_{\mathcal{C}_T}(a, b) = \text{Hom}_{\mathcal{C}}(a, Tb)$,
- The identity morphism on an object a is given by $\eta_a : a \rightarrow Ta$,
- The composition of morphisms $f : a \rightarrow Tb$ and $g : b \rightarrow Tc$ in $\mathcal{C}_T$ is given by the morphism $g \circ_T f : a \rightarrow Tc$ defined as the composition:

$$a \xrightarrow{f} Tb \xrightarrow{Tg} T^2c \xrightarrow{\mu_c} Tc$$

Definition 19 (Extension operator)

Given a monad (T, η, μ) on a category $\mathcal{C}$, the *extension operator* $(-)^{\sharp}$ is defined for each morphism $f : a \rightarrow Tb$ in $\mathcal{C}$ as the morphism $f^{\sharp} : Ta \rightarrow Tb$ given by the composition:

$$Ta \xrightarrow{Tf} T^2b \xrightarrow{\mu_b} Tb$$

In other words, $f^{\sharp} = \mu_b \circ Tf$.

We have the following properties of the extension operator:

- $\eta_a^{\sharp} = \text{id}_{Ta}$ for all objects a in $\mathcal{C}$.
- For any morphism $f : a \rightarrow Tb$, we have $f^{\sharp} \circ \eta_a = f$.
- For any morphisms $f : a \rightarrow Tb$ and $g : b \rightarrow Tc$, we have $(g^{\sharp} \circ f)^{\sharp} = g^{\sharp} \circ f^{\sharp}$.

Definition 20 (Kleisli triple)

A *Kleisli triple* on a category $\mathcal{C}$ consists of:

- A mapping $T : \text{Ob}(\mathcal{C}) \rightarrow \text{Ob}(\mathcal{C})$,
- For each object a in $\mathcal{C}$, a morphism $\eta_a : a \rightarrow Ta$,
- For each morphism $f : a \rightarrow Tb$ in $\mathcal{C}$, a morphism $f^\sharp : Ta \rightarrow Tb$,

such that the following conditions hold:

- $\eta_a^\sharp = \text{id}_{Ta}$ for all objects a in $\mathcal{C}$.
- the following diagrams commute

$$\begin{array}{ccc} a & \xrightarrow{\eta_a} & Ta \\ & \searrow f & \downarrow f^\sharp \\ & & Tb \end{array} \qquad \begin{array}{ccc} Ta & \xrightarrow{f^\sharp} & Tb \\ & \searrow (g^\sharp \cdot f)^\sharp & \downarrow g^\sharp \\ & & Tc \end{array}$$

Proposition 21

Given a category $\mathcal{C}$, there is a one-to-one correspondence between monads on $\mathcal{C}$ and Kleisli triples on $\mathcal{C}$.

Proof

Given a monad (T, η, μ) on $\mathcal{C}$, we can construct a Kleisli triple by defining the extension operator as $f^\sharp = \mu_b \circ Tf$ for each morphism $f : a \rightarrow Tb$.

Conversely, given a Kleisli triple $(T, \eta, (-)^\sharp)$ on $\mathcal{C}$, we can construct a monad by defining the multiplication $\mu_a : T^2a \rightarrow Ta$ as $\mu_a = (\text{id}_{Ta})^\sharp$ for each object a in $\mathcal{C}$.

The conditions of the Kleisli triple ensure that the monad axioms are satisfied, and the construction is reversible, establishing the one-to-one correspondence. $\square$

Remark : Kleisli triples are sometimes referred to as *monads in extensive form*.

Example (Monads in terms of binds and returns)

In the context of programming languages, a monad can be described in terms of a type constructor T and two operations: **return** (or **unit**) and **bind**. The **return** operation corresponds to the unit η of the monad, while the **bind** operation corresponds to the extension operator $(-)^\sharp$. Specifically, for a monad (T, η, μ) on a category $\mathcal{C}$, we can define:

- $\text{return}_a = \eta_a : a \rightarrow Ta$ for each object a in $\mathcal{C}$,
- $\text{bind}_{a,b}(f) = f^\sharp : Ta \rightarrow Tb$ for each morphism $f : a \rightarrow Tb$ in $\mathcal{C}$.

We use the notion of $\mathbb{S}$ -algebra of [MW10] to define the distributive laws for monads in extensive form.

Definition 22 ($\mathbb{S}$ -algebra, [MW10])

Given a monad in extensive form $\mathbb{S} = (S, \eta, (-)^\mathbb{S})$ on a category $\mathcal{C}$, an $\mathbb{S}$ -algebra is a pair $\mathbb{B} = (B, (-)^\mathbb{B})$ where:

- $B \in \mathcal{C}$
- $(-)_X^\mathbb{B} : \mathcal{C}(X, B) \rightarrow \mathcal{C}(SX, B)$

such that the following diagrams commute for every $h : X \rightarrow B$ and $k : Y \rightarrow SX$:

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & SX \\ & \searrow h & \downarrow h^\mathbb{B} \\ & & B \end{array} \quad \begin{array}{ccc} SY & \xrightarrow{k^\mathbb{S}} & SX \\ & \searrow (h^\mathbb{B}k)^\mathbb{B} & \downarrow h^\mathbb{B} \\ & & B \end{array}$$

A morphism of $\mathbb{S}$ -algebras $(B, (-)^\mathbb{B})$ to $(A, (-)^\mathbb{A})$ can be defined as an arrow $l : B \rightarrow A$ in $\mathcal{C}$ such that the following diagram commutes for every $h : X \rightarrow B$:

$$\begin{array}{ccc} SX & \xrightarrow{h^\mathbb{B}} & B \\ & \searrow (l \cdot h)^\mathbb{A} & \downarrow l \\ & & A \end{array}$$

Theorem 23 (Distributive law between monads in extensive form, [MW10])

Let $\mathbb{T} = (T, \eta_T, (-)^\mathbb{T})$ and $\mathbb{S} = (S, \eta_S, (-)^\mathbb{S})$ be monads on $\mathcal{C}$. A distributive law of $\mathbb{T}$ over $\mathbb{S}$ can equivalently be given by an $\mathbb{T}$ -algebra $(STA, (-)^\lambda)$ for every $A \in \mathcal{C}$ such that

- $(S\eta_T A \cdot \eta_S A)^\lambda = \eta_S T A$
- for every $f : B \rightarrow TSA$, $(f^\lambda)^\mathbb{S} : (STB, (-)^\lambda) \rightarrow (STA, (-)^\lambda)$ is a morphism of $\mathbb{T}$ -algebras.

Unfolding the conditions for the $\mathbb{T}$ -algebra and for the morphism, the following diagrams must commute for every $h : X \rightarrow STA$, $k : Y \rightarrow TX$, $f : B \rightarrow STA$ and $g : X \rightarrow STB$:

$$\begin{array}{ccc} X & \xrightarrow{\eta_T X} & TX \\ & \searrow h & \downarrow h^\lambda \\ & & STA \end{array} \quad \begin{array}{ccc} TY & \xrightarrow{k^\mathbb{T}} & TX \\ & \searrow (h^\lambda k)^\lambda & \downarrow h^\lambda \\ & & STA \end{array} \quad \begin{array}{ccc} TB & \xrightarrow{g^\lambda} & STB \\ & \searrow ((f^\lambda)^\mathbb{S} \cdot g)^\lambda & \downarrow (f^\lambda)^\mathbb{S} \\ & & STA \end{array}$$

Proposition 24 ([MW10])

Given a distributive law λ of $\mathbb{T}$ over $\mathbb{S}$, the composite monad is given by $\mathbb{S} \circ_\lambda \mathbb{T} = (ST, S\eta_T \cdot \eta_S, ((-)^\lambda)^\mathbb{S})$

Proposition 25 ([MW10])

Let $\mathbb{S}$ and $\mathbb{T}$ be monads on $\mathcal{C}$. There is a bijection between distributive laws of $\mathbb{T}$ over $\mathbb{S}$ and monad structures $(ST, S\eta_T \cdot \eta_S, (-)^{(\mathbb{ST})})$ on ST such that:

- for every $k : Y \rightarrow TX$, $S(k^{\mathbb{T}}) = (\eta_S TX \cdot k)^{(\mathbb{ST})}$
- for every $m : Y \rightarrow STX$ and $h : X \rightarrow STU$:
 - $h^{(\mathbb{ST})} = (h^{(\mathbb{ST})} \cdot \eta_S TX)^{\mathbb{T}}$
 - the following diagram commutes

$$\begin{array}{ccc} SY & \xrightarrow{m^{\mathbb{S}}} & STX \\ & \searrow (h^{(\mathbb{ST})} \cdot m)^{\mathbb{S}} & \downarrow h^{(\mathbb{ST})} \\ & & STU \end{array}$$

1.4 2-categories**Definition 26 (2-category)**

A 2-category $\mathcal{C}$ consists of:

- Cellular structure:
 - A collection C_0 of 0-cells $a, b, c, \dots$ (objects),
 - For each pair of 0-cells a, b , a collection $C_1(a, b)$ of 1-cells (morphisms) from a to b ,
 - For each pair of 1-cells $f, g \in C_1(a, b)$, a collection $C_2(f, g)$ of 2-cells (2-morphisms) from f to g
- Identities:
 - A 1-cell $\text{id}_a : a \rightarrow a$ for each 0-cell $a \in C_0$ (identity 1-cell),
 - A 2-cell $\text{id}_f : f \Rightarrow f$ for each 1-cell $f \in C_1(a, b)$ (identity 2-cell)
- Composition:
 - A composition operation on 1-cells $g \circ f$ for every pair of composable 1-cells $f : a \rightarrow b$ and $g : b \rightarrow c$,
 - A vertical composition operation on 2-cells $\beta \circ \alpha : f \Rightarrow h$ for every pair of composable 2-cells $\alpha : f \Rightarrow g$ and $\beta : g \Rightarrow h$,
 - A horizontal composition operation on 2-cells $\beta \bullet \alpha : g \circ f \Rightarrow g' \circ f'$ for every pair of composable 2-cells $\alpha : f \Rightarrow f'$ and $\beta : g \Rightarrow g'$

such that

- 1-cell compositions are associative and identities are neutral
- 2-cell vertical compositions are associative and identities are neutral
- 2-cell horizontal compositions are associative and identities are neutral, and composition of 2-cell identities preserves 1-cell composition
- the interchange law holds: for any composable 2-cells $\alpha, \alpha', \beta, \beta'$, we have

$$(\beta' \circ \beta) \bullet (\alpha' \circ \alpha) = (\beta' \bullet \alpha') \circ (\beta \bullet \alpha)$$

Example (2-category of Categories)

The 2-category **Cat** has:

- 0-cells: categories,
- 1-cells: functors between categories,
- 2-cells: natural transformations between functors.
- pointwise composition of functors as 1-cell composition,
- pointwise composition of natural transformations as vertical composition,
- horizontal composition of natural transformations defined by the composition of functors.

Definition 27 (Adjunction in a 2-category)

Given a 2-category C , an *adjunction* between 0-cells a and b in C consists of:

- Two 1-cells $f : a \rightarrow b$ and $g : b \rightarrow a$,
- Two 2-cells $\eta : \text{id}_a \Rightarrow g \circ f$ (unit) and $\varepsilon : f \circ g \Rightarrow \text{id}_b$ (counit),

such that the following triangle identities hold:

$$(f \bullet \eta) \circ (\varepsilon \bullet f) = \text{id}_f \quad (\eta \bullet g) \circ (g \bullet \varepsilon) = \text{id}_g$$

1.5 Multicategories**Definition 28 (Multicategory)**

A multicategory $\mathcal{U}$ consists of:

- A class of objects $\text{Ob}(\mathcal{U})$,
- For each finite (possibly empty) sequence of objects $(a_1, a_2, \dots, a_n)$ and an object b , a set of morphisms $\mathcal{U}(a_1, a_2, \dots, a_n; b)$,
- For each object a , an identity morphism $\text{id}_a : (a) \rightarrow a$,
- A composition operation that allows for the composition of morphisms in a way that generalizes the composition in a category, satisfying appropriate associativity and identity axioms.

We denote by **Multicat** the set of all multicategories.

Example (Category with Finite Products as a Multicategory)

Given a category $\mathcal{C}$ with finite products, we can construct a multicategory $\mathcal{U}$ where:

- The objects of $\mathcal{U}$ are the same as the objects of $\mathcal{C}$,
- The morphisms $\mathcal{U}(a_1, a_2, \dots, a_n; b)$ are given by the morphisms in $\mathcal{C}$ from the product $a_1 \times a_2 \times \dots \times a_n$ to b ,
- The identities rely on the canonical isomorphism

$$\text{id}_a := \prod_{i=1}^n a \cong a$$

- The composition is defined using the universal property of products and the composition in $\mathcal{C}$.

Example (Monoidal Category as a Multicategory)

A monoidal category $(\mathcal{C}, \otimes, I)$ can be viewed as a multicategory where:

- The objects of the multicategory are the same as the objects of $\mathcal{C}$,
- The morphisms $\mathcal{U}(a_1, a_2, \dots, a_n; b)$ are given by the morphisms in $\mathcal{C}$ from $a_1 \otimes a_2 \otimes \dots \otimes a_n$ to b (by nesting the monoidal product)

Definition 29 (Multifunctor)

A *multi-functor* $F : \mathcal{U} \rightarrow \mathcal{V}$ between multicategories $\mathcal{U}$ and $\mathcal{V}$ consists of two mappings:

- A mapping $F : \text{Ob}(\mathcal{U}) \rightarrow \text{Ob}(\mathcal{V})$ that assigns to each object a in $\mathcal{U}$ an object Fa in $\mathcal{V}$,
- A mapping $F : \mathcal{U}(a_1, a_2, \dots, a_n; b) \rightarrow \mathcal{V}(Fa_1, Fa_2, \dots, Fa_n; Fb)$ that assigns to each morphism $f : (a_1, a_2, \dots, a_n) \rightarrow b$ in $\mathcal{U}$ a morphism $Ff : (Fa_1, Fa_2, \dots, Fa_n) \rightarrow Fb$ in $\mathcal{V}$,

such that F preserves:

- Identities: For each object a in $\mathcal{U}$, $F\text{id}_a = \text{id}_{Fa}$,
- Composition: For morphisms $f : (a_1, a_2, \dots, a_n) \rightarrow b$ and $g_i : (a_{i1}, a_{i2}, \dots, a_{ik_i}) \rightarrow a_i$ in $\mathcal{U}$, we have

$$F(f \circ (g_1, g_2, \dots, g_n)) = Ff \circ (Fg_1, Fg_2, \dots, Fg_n).$$

Definition 30 (Multi-natural transformation)

Given two multi-functors $F, G : \mathcal{U} \rightarrow \mathcal{V}$ between multicategories $\mathcal{U}$ and $\mathcal{V}$, a *multi-natural transformation* $\eta : F \rightarrow G$ consists of a family of morphisms $\{\eta_a : Fa \rightarrow Ga\}_{a \in \text{Ob}(\mathcal{U})}$ such that for every morphism $f : (a_1, a_2, \dots, a_n) \rightarrow b$ in $\mathcal{U}$, the following diagram commutes:

$$\begin{array}{ccc} (Fa_1, Fa_2, \dots, Fa_n) & \xrightarrow{Ff} & Fb \\ (\eta_{a_1}, \eta_{a_2}, \dots, \eta_{a_n}) \downarrow & & \downarrow \eta_b \\ (Ga_1, Ga_2, \dots, Ga_n) & \xrightarrow{Gf} & Gb \end{array}$$

Definition 31 (Model)

Let $\mathcal{O}$ be an operad, and $\mathcal{U}$ be a multicategory. A *model* of $\mathcal{O}$ in $\mathcal{U}$ is a multi-functor $\mathcal{A} : \mathcal{O} \rightarrow \mathcal{U}$ that preserves the operadic structure, i.e., it respects the composition and identity of the operad.

Definition 32

Homomorphism of Models A *homomorphism* $h : \mathcal{A} \rightarrow \mathcal{B}$ between $\mathcal{O}$ -models in $\mathcal{U}$ consists of a morphism $\underline{h} : (\underline{\mathcal{A}}) \rightarrow (\underline{\mathcal{B}})$ such that for all $f \in \mathcal{O}_n$,

$$\underline{h} \circ (\mathcal{A}[[f]]) = \mathcal{B}[[f]](\underline{h}, \dots, \underline{h})$$

Example (Category of Models)

Given an operad $\mathcal{O}$ and a multicategory $\mathcal{U}$, we can construct a category $\mathbf{Mod}(\mathcal{O}, \mathcal{U})$ where:

- The objects are the models of $\mathcal{O}$ in $\mathcal{U}$,
- The morphisms are the homomorphisms between these models.

1.6 Free algebras

We'll use the definitions from [All+25].

Definition 33 (Finitary signature)

A *finitary signature* $\Sigma = (\text{op } \Sigma, \text{arity})$ consists of a set $\text{op } \Sigma$ of *operators*, and an assignment $\text{arity} : \text{op } \Sigma \rightarrow \mathbb{N}$ associating to each operator in $\text{op } \Sigma$ its *arity*.

Definition 34 (Algebra)

An *algebra* $A = (\underline{A}, A[[\text{--}]])$ over a finitary signature Σ consists of:

- A set $\underline{A}$ called the *carrier* of A ,
- For each operator $f \in \text{op } \Sigma$ of arity n , a function $A[[f]] : \underline{A}^n \rightarrow \underline{A}$ called the *interpretation* of f in A .

Definition 35 (Terms)

Given a set X of variables and a finitary signature Σ , the set of *terms* $\text{Term}_\Sigma(X)$ is defined inductively as follows:

- Every variable $x \in X$ is a term,
- If $f \in \text{op } \Sigma$ is an operator of arity n and $t_1, t_2, \dots, t_n$ are terms, then $f(t_1, t_2, \dots, t_n)$ is a term.

Remark : The definition of algebraic theory seen before can be reformulated as a finitary signature together with a set of equations between terms built from the operators in the signature.

Definition 36 (Environment and Interpretation)

An *environment* ρ for a context X in an algebra A is a function $\rho : X \rightarrow \underline{A}$ that assigns to each variable in X an element of the carrier of A .

Given a term $t \in \text{Term}_\Sigma(X)$, denoted $X \vdash t$, the algebra A gives an *interpretation* function $A[[t]] : \underline{A}^X \rightarrow \underline{A}$. Given an environment $\rho : X \rightarrow \underline{A}$, we can evaluate the term t in A under the environment ρ , denoted $A[[t]](\rho)$, by recursively substituting the variables in t with their corresponding values in $\underline{A}$ according to ρ and applying the interpretations of the operators as defined in A .

Definition 37 (Valid equations)

An equation $t = t'$ between terms in $\text{Term}_\Sigma(X)$ is *valid* in an algebra A if for every environment $\rho : X \rightarrow \underline{A}$, we have $A[[t]](\rho) = A[[t']](\rho)$.

Definition 38 (Presentation and Models)

A *presentation* $\mathcal{T} = (\Sigma_{\mathcal{T}}, \mathcal{T}_{ax})$ consists of a finitary signature $\Sigma_{\mathcal{T}}$ and a set $\mathcal{T}_{ax}$ of *axioms*, which are equations between terms in $\text{Term}_{\Sigma_{\mathcal{T}}}(X)$ for some set of variables X .

A $\mathcal{T}$ -model A is a $\Sigma_{\mathcal{T}}$ -algebra such that all equations in $\mathcal{T}_{ax}$ are valid in A .

A $\mathcal{T}$ -model A over a context X is a $\mathcal{T}$ -model A together with an environment $\rho : X \rightarrow \underline{A}$.

Definition 39 (Homomorphisms)

Let $A = (\underline{A}, A[\![-]\!])$ and $B = (\underline{B}, B[\![-]\!])$ be algebras over the same signature Σ .

A *homomorphism* of Σ -algebras from A to B is a function $h : \underline{A} \rightarrow \underline{B}$ such that for every operator $f \in \text{op } \Sigma$ of arity n , we have:

$$B[\![f]\!](h(a_1), \dots, h(a_n)) = h(A[\![f]\!](a_1, \dots, a_n))$$

for all $a_1, \dots, a_n \in \underline{A}$.

A *homomorphism* of $\mathcal{T}$ -models is a homomorphism of $\Sigma_{\mathcal{T}}$ -algebras between the underlying algebras of the models.

Definition 40 (Morphism)

A *morphism* $h : A \rightarrow B$ of $\mathcal{T}$ -models over X is a $\mathcal{T}$ -model homomorphism that makes the diagram commute:

$$\begin{array}{ccc} X & \xrightarrow{\rho_A} & \underline{A} \\ & \searrow \rho_B & \downarrow h \\ & & \underline{B} \end{array}$$

Definition 41 (Free model (or free algebra))

Let $\mathcal{T}$ be a presentation and X be a set of variables.

A *free $\mathcal{T}$ -model* over X is a $\mathcal{T}$ -model $F(X)$ such that for any $\mathcal{T}$ -model A over X , there exists a unique morphism of $\mathcal{T}$ -models from $F(X)$ to A .

The existence-and-uniqueness property is called the *universal property* of free $\mathcal{T}$ -models over X .

Informally:

- A signature gives defines the operators and constants (arity 0 operators).
- An algebra gives a concrete interpretation of the signature.
- A presentation defines axioms on top of the signature, for instance the axioms of a group, or of a semiring, etc.
- A model of a presentation is an algebra that satisfies the axioms of the presentation.
- A free algebra is the most general model of a presentation.

We can give an equivalent definition for free algebras using universal arrows.

Proposition 42

Let $\mathcal{T}$ be a presentation, and X be a set. Let $\mathcal{T}\text{-}\mathbf{Mod}$ be the category of $\mathcal{T}$ -models over X . Let $U : \mathcal{T}\text{-}\mathbf{Mod} \rightarrow \mathbf{Set}$ be the forgetful functor that sends a $\mathcal{T}$ -model to its underlying set.

Then, a free $\mathcal{T}$ -model over X is a universal arrow from X to U .

Proof

Let $F X$ be a free $\mathcal{T}$ -model over X . We want to show that $(F X, \eta_X)$ is a universal arrow from X to U , where $\eta_X : X \rightarrow U F X$ is the function that sends each variable in X to its interpretation in $F X$.

Let A be a $\mathcal{T}$ -model over X , and let $\rho_A : X \rightarrow U A$ be the environment of A . Since $F X$ is a free $\mathcal{T}$ -model over X , there exists a unique morphism of $\mathcal{T}$ -models $h : F X \rightarrow A$ such that the following

diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & UF X \\ & \searrow \rho_A & \downarrow U h \\ & & U A \end{array}$$

This shows that $(F X, \eta_X)$ satisfies the universal property of being a universal arrow from X to U .

Conversely, if $(F X, \eta_X)$ is a universal arrow from X to U , then for any $\mathcal{T}$ -model A over X , there exists a unique morphism of $\mathcal{T}$ -models from $F X$ to A , which means that $F X$ is a free $\mathcal{T}$ -model over X . Therefore, the two definitions are equivalent. $\square$

2 SOME FREE ALGEBRAS

2.1 Semigroups

Free commutative semigroups

Let's try to construct the free algebra for the presentation of commutative semigroups. We cannot rely on the order of the terms, nor the order of applying the operations.

Let X be a set. We consider the theory of commutative semigroups **CSG**, with a signature Σ with a single operator $(+)$ of arity 2, and with the two axioms

$$\begin{aligned} a, b &\vdash a + b = b + a \\ a, b, c &\vdash a + (b + c) = (a + b) + c \end{aligned}$$

We pose

$$SX = \{f : X \rightarrow \mathbb{N} \mid f \text{ has finite non-empty support}\}$$

For $f, g \in SX$, we pose

$$f + g = \lambda x. f(x) + g(x) \quad x \in X$$

We define $\eta_{\text{CSG}} : X \rightarrow SX$ for all $x \in X$ by

$$\eta_{\text{CSG}}(x) = \lambda y. \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise} \end{cases}$$

We pose $F_{\text{CSG}} X = (SX, (+))$ where $+$ is the operator defined above.

Proposition 43

$F_{\text{CSG}} X$ is a free **CSG**-model over X .

Proof

- ▷ Let's first check that $F_{\text{CSG}} X$ is indeed a **CSG**-model. We need to check that the axioms of **CSG** are valid in $F_{\text{CSG}} X$. This is straightforward and comes down to the commutativity and associativity of addition in $\mathbb{N}$.
- ▷ Let A be a **CSG**-model over X with environment $\varphi_A : X \rightarrow U A$. We want to show that there exists a unique morphism h of **CSG**-models from $F_{\text{CSG}} X$ to A such that the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{\eta_{\text{CSG}}} & SX \\ & \searrow \varphi_A & \downarrow h \\ & & U A \end{array}$$

with U being the forgetful functor from **CSG**-models to **Set**.

We define $\Sigma\varphi_A : SX \rightarrow A$ for all $f \in SX$ by

$$\Sigma\varphi_A(f) = \sum_{x \in X} f \ x \cdot \varphi_A(x)$$

We want to show that $\Sigma\varphi_A$ is the unique morphism h of **CSG**-models from $F_{\mathbf{CSG}}X$ to A above. We need to check that $\Sigma\varphi_A$ preserves the operator $+$, and that the diagram above commutes.

- Preservation of the operator $+$: let $f, g \in SX$. We have:

$$\begin{aligned} \Sigma\varphi_A(f + g) &= \Sigma\varphi_A(\lambda x. f \ x + g \ x) \\ &= \sum_{x \in X} (f \ x + g \ x) \cdot \varphi_A(x) \\ &= \sum_{x \in X} f \ x \cdot \varphi_A(x) + \sum_{x \in X} g \ x \cdot \varphi_A(x) \\ &= \Sigma\varphi_A(f) + \Sigma\varphi_A(g) \end{aligned}$$

- Commutation of the diagram: let $x \in X$. We have:

$$\begin{aligned} (\Sigma\varphi_A)(\eta_{\mathbf{CSG}}(x)) &= \Sigma\varphi_A \left(\lambda y. \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise} \end{cases} \right) \\ &= \sum_{y \in X} \begin{cases} 1 \cdot \varphi_A(x) & \text{if } x = y \\ 0 & \text{otherwise} \end{cases} \\ &= \varphi_A(x) \end{aligned}$$

Therefore, the diagram commutes.

- Uniqueness: let $h : F_{\mathbf{CSG}}X \rightarrow A$ be a morphism of **CSG**-models such that the diagram above commutes. We want to show that $h = \Sigma\varphi_A$. Let $f \in SX$. We have:

$$\begin{aligned} h(f) &= h \left(\sum_{x \in X} f \ x \cdot \eta_{\mathbf{CSG}}(x) \right) \\ &= \sum_{x \in X} f \ x \cdot h(\eta_{\mathbf{CSG}}(x)) \\ &= \sum_{x \in X} f \ x \cdot \varphi_A(x) && \text{since the diagram commutes} \\ &= \Sigma\varphi_A(f) \end{aligned}$$

Therefore, we proved that $F_{\mathbf{CSG}}X$ is a free **CSG**-model over X .

□

Finally we'll define a multiplication $\mu_{\mathbf{CSG}}$ so that we have a monad structure $(S, \eta_{\mathbf{CSG}}, \mu_{\mathbf{CSG}})$ on **Set**. For all $f \in SSX$, we pose

$$\mu_{\mathbf{CSG}}(f) = \lambda x. \sum_{g \in SX} f \ g \cdot g \ x$$

Proposition 44

$(S, \eta_{\mathbf{CSG}}, \mu_{\mathbf{CSG}})$ is a monad on **Set**.

Proof

We need to check the monad axioms:

- Left unit: let $f \in SX$. We have:

$$\begin{aligned}
 \mu_{\mathbf{CSG}}(\eta_{\mathbf{CSG}}S(f)) &= \mu_{\mathbf{CSG}}\left(\lambda g. \begin{cases} 1 & \text{if } g = f \\ 0 & \text{otherwise} \end{cases}\right) \\
 &= \lambda x. \sum_{g \in SX} \begin{cases} 1 \cdot g & \text{if } g = f \\ 0 & \text{otherwise} \end{cases} \\
 &= \lambda x. f \cdot x \\
 &= f
 \end{aligned}$$

- Right unit: let $f \in SX$. We have:

$$\begin{aligned}
 \mu_{\mathbf{CSG}}(S\eta_{\mathbf{CSG}}(f)) &= \mu_{\mathbf{CSG}}\left(\lambda g. \begin{cases} f \cdot x & \text{if } g = \eta_{\mathbf{CSG}}(x) \text{ for some } x \in X \\ 0 & \text{otherwise} \end{cases}\right) \\
 &= \lambda x. \sum_{g \in SX} \begin{cases} f \cdot x \cdot g & \text{if } g = \eta_{\mathbf{CSG}}(x) \text{ for some } x \in X \\ 0 & \text{otherwise} \end{cases} \\
 &= \lambda x. f(x) \cdot \eta_{\mathbf{CSG}}(x)(x) \\
 &= \lambda x. f \cdot x \\
 &= f
 \end{aligned}$$

- Associativity: let $f \in SSSX$. We have:

$$\begin{aligned}
 \mu_{\mathbf{CSG}}(S\mu_{\mathbf{CSG}}(f)) &= \mu_{\mathbf{CSG}}\left(\lambda g. \sum_{h \in SSSX} f \cdot h \cdot h \cdot g\right) \\
 &= \lambda x. \sum_{g \in SX} \left(\sum_{h \in SSSX} f \cdot h \cdot h \cdot g \right) \cdot g \cdot x \\
 &= \lambda x. \sum_{h \in SSSX} f \cdot h \cdot \left(\sum_{g \in SX} h \cdot g \cdot g \cdot x \right) \\
 &= \lambda x. \sum_{h \in SSSX} f \cdot h \cdot \mu_{\mathbf{CSG}}(h) \\
 &= \mu_{\mathbf{CSG}}(\mu_{\mathbf{CSG}}S(f))
 \end{aligned}$$

□

Free semigroups

Compared to the free commutative semigroup, we can rely on the order of the terms. We'll use lists to represent the terms of the free semigroup.

Let X be a set. We consider the theory of semigroups $\mathbf{SG}$, with a signature Σ with a single operator $(\cdot)$ of arity 2, and with the axiom

$$a, b, c \vdash a \cdot (b \cdot c) = (a \cdot b) \cdot c$$

We pose

$$TX = \{l \in X^* \mid l \text{ is finite and non-empty}\}$$

For $f, g \in TX$, we pose

$$f \cdot g = f \mathbin{++} g$$

where $++$ is the concatenation of lists.

We define $\eta_{\mathbf{SG}} : X \rightarrow TX$ for all $x \in X$ by

$$\eta_{\mathbf{SG}}(x) = [x]$$

We pose $F_{\mathbf{SG}}X = (TX, (\cdot))$ where $\cdot$ is the operator defined above.

Proposition 45

$F_{\mathbf{SG}}X$ is a free $\mathbf{SG}$ -model over X .

Proof

▷ $F_{\mathbf{SG}}$ is a $\mathbf{SG}$ -model: the associativity of $\cdot$ in $F_{\mathbf{SG}}$ comes from the associativity of list concatenation.

▷ Let A be a $\mathbf{SG}$ -model over X with environment $\varphi_A : X \rightarrow U A$.

We define $\Sigma\varphi_A : TX \rightarrow \underline{A}$ for all $l \in TX$ by

$$\Sigma\varphi_A(l) = \prod_{x \in l} \varphi_A(x)$$

where the product^a is taken in the order of the list l , which is well defined because the list is finite, non-empty, and the product is associative.

Let's show that $\Sigma\varphi_A$ is the unique morphism h of $\mathbf{SG}$ -models from $F_{\mathbf{SG}}X$ to A such that the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{\eta_{\mathbf{SG}}} & TX \\ & \searrow \varphi_A & \downarrow h \\ & & U A \end{array}$$

– Preservation of the operator $\cdot$: let $f, g \in TX$. We have:

$$\begin{aligned} \Sigma\varphi_A(f \cdot g) &= \Sigma\varphi_A(f \mathbin{++} g) \\ &= \prod_{x \in f \mathbin{++} g} \varphi_A(x) \\ &= \prod_{x \in f} \varphi_A(x) \cdot \prod_{x \in g} \varphi_A(x) && \text{by associativity of the product} \\ &= \Sigma\varphi_A(f) \cdot \Sigma\varphi_A(g) \end{aligned}$$

– Commutation of the diagram: let $x \in X$. We have:

$$\begin{aligned} (\Sigma\varphi_A)(\eta_{\mathbf{SG}}(x)) &= \Sigma\varphi_A([x]) \\ &= \varphi_A(x) \end{aligned}$$

– Uniqueness: let $h : F_{\mathbf{SG}}X \rightarrow A$ be a morphism of $\mathbf{SG}$ -models such that the diagram above commutes. We want to show that $h = \Sigma\varphi_A$. Let $l \in TX$. We have:

$$\begin{aligned} h(l) &= h\left(\prod_{x \in l} \eta_{\mathbf{SG}}(x)\right) \\ &= \prod_{x \in l} h(\eta_{\mathbf{SG}}(x)) && \text{since } h \text{ is a morphism} \\ &= \prod_{x \in l} \varphi_A(x) && \text{since the diagram commutes} \\ &= \Sigma\varphi_A(l) \end{aligned}$$

Therefore we proved that $F_{\mathbf{SG}}X$ is a free $\mathbf{SG}$ -model over X .

□

^aproduct with the operator $\cdot$

We'll also define a multiplication $\mu_{\mathbf{SG}}$ so that we have a monad structure $(T, \eta_{\mathbf{SG}}, \mu_{\mathbf{SG}})$ on $\mathbf{Set}$. For all $l \in TTX$, we pose

$$\mu_{\mathbf{SG}}(l) = \mathrel{+}_{x \in l} x$$

where the concatenation is taken in the order of the list l , which is well defined because the list l is finite, non-empty, and the concatenation is associative.

Proposition 46

$(T, \eta_{\mathbf{SG}}, \mu_{\mathbf{SG}})$ is a monad on $\mathbf{Set}$.

Proof

We need to check the monad axioms:

- Left unit: let $l \in TX$. We have:

$$\begin{aligned} \mu_{\mathbf{SG}}(\eta_{\mathbf{SG}}T(l)) &= \mu_{\mathbf{SG}}([l]) \\ &= \mathrel{+}_{x \in [l]} x \\ &= l \end{aligned}$$

- Right unit: let $l \in TX$. We have:

$$\begin{aligned} \mu_{\mathbf{SG}}(T\eta_{\mathbf{SG}}(l)) &= \mu_{\mathbf{SG}}([[x_i] \mid x_i \in l]) \\ &= \mathrel{+}_{x \in [[x_i] \mid x_i \in l]} x \\ &= \mathrel{+}_{x_i \in l} [x_i] \\ &= l \end{aligned}$$

- Associativity: let $l \in TTTX$. We have:

$$\begin{aligned} \mu_{\mathbf{SG}}(T\mu_{\mathbf{SG}}(l)) &= \mu_{\mathbf{SG}}([\mathrel{+}_{x \in l_i} x \mid l_i \in l]) \\ &= \mathrel{+}_{x \in [\mathrel{+}_{x \in l_i} x \mid l_i \in l]} x \\ &= \mathrel{+}_{l_i \in l} \mathrel{+}_{x \in l_i} x \\ &= \mathrel{+}_{x \in \mathrel{+}_{l_j \in l} l_j} x \\ &= \mathrel{+}_{x \in l} x \\ &= \mu_{\mathbf{SG}}(\mu_{\mathbf{SG}}T(l)) \end{aligned}$$

□

2.2 A first combination

There is no name for a presentation composed of a semigroup for the product and a commutative semigroup for the sum, so we'll just call it $\mathbf{SG} + \mathbf{CSG}$ for now.

The signature of $\mathbf{SG} + \mathbf{CSG}$ has two operators: $(\cdot)$ of arity 2, and $(+)$ of arity 2, and the axioms are the union of the axioms of $\mathbf{SG}$ and $\mathbf{CSG}$:

$$\begin{aligned} a, b, c &\vdash a \cdot (b \cdot c) = (a \cdot b) \cdot c \\ a, b &\vdash a + b = b + a \\ a, b, c &\vdash a + (b + c) = (a + b) + c \end{aligned}$$

We'll also add the distributivity axioms:

$$\begin{aligned} a, b, c &\vdash a \cdot (b + c) = a \cdot b + a \cdot c && \text{left distributivity} \\ a, b, c &\vdash (a + b) \cdot c = a \cdot c + b \cdot c && \text{right distributivity} \end{aligned}$$

We'll try to combine the two previous examples to hopefully get the free algebra for the presentation.

The carrier set of our candidate for the free **SG** + **CSG**-model over X is STX .

We will keep the same definition for the operator $+$ as in the free commutative semigroup, but we'll need to change the definition of the operator $\cdot$ to take into account the fact that we are now working with STX instead of TX .

If we write $(p : n) \in f$ when $f(p) = n$, then what we want is

$$f \cdot g = \sum_{(p:n) \in f} \sum_{(q:m) \in g} (p \cdot_{\mathbf{SG}} q : n \cdot m)$$

Formally, this can be defined for all $f, g \in STX$ as

$$f \cdot g = \lambda l : TX. \sum_{l = l_1 \cdot_{\mathbf{SG}} l_2} f(l_1) \cdot g(l_2)$$

We define $\eta_{\mathbf{SG}+\mathbf{CSG}} : X \rightarrow STX = \eta_{\mathbf{CSG}} \circ \eta_{\mathbf{SG}}$. Let's first prove that the following diagram commutes:

$$\begin{array}{ccccc} X & \xrightarrow{\eta_{\mathbf{SG}}} & TX & \xrightarrow{\eta_{\mathbf{CSG}}^T} & STX \\ & \searrow \eta_{\mathbf{CSG}} & & \nearrow S\eta_{\mathbf{SG}} & \\ & & SX & & \end{array}$$

Let $x \in X$. We have:

$$\begin{aligned} (S\eta_{\mathbf{SG}})(\eta_{\mathbf{CSG}}(x)) &= S\eta_{\mathbf{SG}} \left(\lambda y. \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise} \end{cases} \right) \\ &= \lambda y. \begin{cases} 1 & \text{if } \eta_{\mathbf{SG}}(x) = y \\ 0 & \text{otherwise} \end{cases} \\ &= \eta_{\mathbf{CSG}}(\eta_{\mathbf{SG}}(x)) \end{aligned}$$

We pose $F_{\mathbf{SG}+\mathbf{CSG}}X = (STX, (\cdot), (+))$, where $(\cdot)$ and $(+)$ are defined above.

Proposition 47

$F_{\mathbf{SG}+\mathbf{CSG}}X$ is a free **SG** + **CSG**-model over X .

Proof

▷ Let's first check that $F_{\mathbf{SG}+\mathbf{CSG}}X$ is indeed a **SG** + **CSG**-model.

The associativity and commutativity of $(+)$ was already proven for $F_{\mathbf{CSG}}X$.

For the associativity of $(\cdot)$: let $f, g, h \in STX$. We have:

$$\begin{aligned}
(f \cdot g) \cdot h &= \lambda l : TX. \sum_{l=l_1 \cdot \mathbf{SG} l_2} (f \cdot g)(l_1) \cdot h(l_2) \\
&= \lambda l : TX. \sum_{l=l_1 \cdot \mathbf{SG} l_2} \sum_{l_1=l_{11} \cdot \mathbf{SG} l_{12}} f(l_{11}) \cdot g(l_{12}) \cdot h(l_2) \\
&= \lambda l : TX. \sum_{l=l_{11} \cdot \mathbf{SG} (l_{12} \cdot \mathbf{SG} l_2)} f(l_{11}) \cdot g(l_{12}) \cdot h(l_2) && \text{by associativity of } \cdot \mathbf{SG} \\
&= \lambda l : TX. \sum_{l=l_{11} \cdot \mathbf{SG} l_3} \sum_{l_3=l_{12} \cdot \mathbf{SG} l_2} f(l_{11}) \cdot g(l_{12}) \cdot h(l_2) \\
&= \lambda l : TX. \sum_{l=l_{11} \cdot \mathbf{SG} l_3} f(l_{11}) \cdot (g \cdot h)(l_3) \\
&= f \cdot (g \cdot h)
\end{aligned}$$

The distributivity of $(\cdot)$ over $(+)$ comes from the distributivity of the product in $\mathbb{N}$ over addition, and from the definition of $(+)$.

Therefore, $F_{\mathbf{SG}+\mathbf{CSG}}X$ is a $\mathbf{SG} + \mathbf{CSG}$ -model.

- ▷ Let A be a $\mathbf{SG} + \mathbf{CSG}$ -model over X with environment $\varphi_A : X \rightarrow U A$. We want to show that there exists a unique morphism h of $\mathbf{SG} + \mathbf{CSG}$ -models from $F_{\mathbf{SG}+\mathbf{CSG}}X$ to A such that the following diagram commutes:

$$\begin{array}{ccc}
X & \xrightarrow{\eta_{\mathbf{SG}+\mathbf{CSG}}} & STX \\
& \searrow \varphi_A & \downarrow h \\
& & UA
\end{array}$$

We define $\Sigma\varphi_A : STX \rightarrow \underline{A}$ for all $f \in STX$ by

$$\Sigma\varphi_A(f) = \sum_{l \in TX} f \cdot l \cdot \prod_{x \in l} \varphi_A(x)$$

Let's show that $\Sigma\varphi_A$ is the unique morphism h of $\mathbf{SG} + \mathbf{CSG}$ -models from $F_{\mathbf{SG}+\mathbf{CSG}}X$ to A above. We need to check that $\Sigma\varphi_A$ preserves the operators $(\cdot)$ and $(+)$, and that the diagram above commutes.

- Preservation of the operator $(\cdot)$: let $f, g \in STX$. We have:

$$\begin{aligned}
\Sigma\varphi_A(f \cdot g) &= \Sigma\varphi_A(\lambda l : TX. \sum_{l=l_1 \cdot \mathbf{SG} l_2} f(l_1) \cdot g(l_2)) \\
&= \sum_{l \in TX} \left(\sum_{l=l_1 \cdot \mathbf{SG} l_2} f(l_1) \cdot g(l_2) \right) \cdot \prod_{x \in l} \varphi_A(x) \\
&= \sum_{l_1, l_2 \in TX} f(l_1) \cdot g(l_2) \cdot \prod_{x \in l_1 \cdot \mathbf{SG} l_2} \varphi_A(x) \\
&= \sum_{l_1, l_2 \in TX} f(l_1) \cdot g(l_2) \cdot \prod_{x \in l_1} \varphi_A(x) \cdot \prod_{x \in l_2} \varphi_A(x) && \text{by associativity of the product} \\
&= \sum_{l_1 \in TX} f(l_1) \cdot \prod_{x \in l_1} \varphi_A(x) \cdot \sum_{l_2 \in TX} g(l_2) \cdot \prod_{x \in l_2} \varphi_A(x) \\
&= \Sigma\varphi_A(f) \cdot \Sigma\varphi_A(g)
\end{aligned}$$

- Preservation of the operator (+): let $f, g \in STX$. We have:

$$\begin{aligned}
 \Sigma\varphi_A(f + g) &= \Sigma\varphi_A(\lambda l : TX. f \ l + g \ l) \\
 &= \sum_{l \in TX} (f \ l + g \ l) \cdot \prod_{x \in l} \varphi_A(x) \\
 &= \sum_{l \in TX} f \ l \cdot \prod_{x \in l} \varphi_A(x) + \sum_{l \in TX} g \ l \cdot \prod_{x \in l} \varphi_A(x) \\
 &= \Sigma\varphi_A(f) + \Sigma\varphi_A(g)
 \end{aligned}$$

- Commutation of the diagram: let $x \in X$. We have:

$$\begin{aligned}
 (\Sigma\varphi_A)(\eta_{\mathbf{SG}+\mathbf{CSG}}(x)) &= \Sigma\varphi_A(\eta_{\mathbf{CSG}}(\eta_{\mathbf{SG}}(x))) \\
 &= \Sigma\varphi_A\left(\lambda y. \begin{cases} 1 & \text{if } \eta_{\mathbf{SG}}(x) = y \\ 0 & \text{otherwise} \end{cases}\right) \\
 &= \sum_{y \in TX} \begin{cases} 1 \cdot \prod_{x' \in y} \varphi_A(x') & \text{if } \eta_{\mathbf{SG}}(x) = y \\ 0 & \text{otherwise} \end{cases} \\
 &= \prod_{x' \in \eta_{\mathbf{SG}}(x)} \varphi_A(x') \\
 &= \varphi_A(x)
 \end{aligned}$$

- Uniqueness: let $h : F_{\mathbf{SG}+\mathbf{CSG}}X \rightarrow A$ be a morphism of $\mathbf{SG} + \mathbf{CSG}$ -models such that the diagram above commutes. We want to show that $h = \Sigma\varphi_A$. Let $f \in STX$. We have:

$$\begin{aligned}
 h(f) &= h\left(\sum_{l \in TX} f \ l \cdot \prod_{x \in l} \eta_{\mathbf{SG}}(x)\right) \\
 &= \sum_{l \in TX} f \ l \cdot \prod_{x \in l} h(\eta_{\mathbf{SG}}(x)) && \text{since } h \text{ is a morphism} \\
 &= \sum_{l \in TX} f \ l \cdot \prod_{x \in l} \varphi_A(x) && \text{since the diagram commutes} \\
 &= \Sigma\varphi_A(f)
 \end{aligned}$$

Therefore $F_{\mathbf{SG}+\mathbf{CSG}}X$ is a free $\mathbf{SG} + \mathbf{CSG}$ -model over X .

□

We also define a distributive law $\lambda : TSX \rightarrow STX$ for all $l \in TSX$ by

$$\lambda([f_1, \dots, f_n]) = \sum_{(x_1:n_1) \in f_1} \cdots \sum_{(x_n:n_n) \in f_n} \left([x_1, \dots, x_n] : \prod_{i=1}^n n_i \right)$$

Formaly, this can be defined for all $l \in TSX$ as

$$\lambda(l) = \lambda x : TX. \begin{cases} 0 & \text{if } |x| \neq |l| \\ \prod_{(y,f) \in x * l} f(y) & \text{otherwise} \end{cases}$$

where $x * l$ is the elementwise pairing of the list x and the list l , which is well defined because x and l have the same length.

Proposition 48

λ is a distributive law of the monad T over the monad S .

Proof

We need to check the following axioms:

- $\lambda \circ T\eta_S = \eta_S T$: let $l \in TX$. We have:

$$\begin{aligned}
(\lambda \circ T\eta_S)(l) &= \lambda(T\eta_S(l)) \\
&= \lambda([\eta_S(x) \mid x \in l]) \\
&= \lambda x : TX. \begin{cases} 0 & \text{if } |x| \neq |l| \\ \prod_{(y,f) \in x * [\eta_S(x) \mid x \in l]} f(y) & \text{otherwise} \end{cases} \\
&= \lambda x : TX. \begin{cases} 0 & \text{if } |x| \neq |l| \\ \prod_{(y,z) \in x * l} \eta_S(z)(y) & \text{otherwise} \end{cases} \\
&= \eta_S T(l)
\end{aligned}$$

- $\lambda \circ \eta_T S = S\eta_T$: let $f \in SX$. We have:

$$\begin{aligned}
(\lambda \circ \eta_T S)(f) &= \lambda(\eta_T S(f)) \\
&= \lambda([f]) \\
&= \lambda x : TX. \begin{cases} 0 & \text{if } |x| \neq 1 \\ \prod_{(y,g) \in x * [f]} g(y) & \text{otherwise} \end{cases} \\
&= S\eta_T(f)
\end{aligned}$$

- $\lambda \circ T\mu_S = \mu_S T \circ S\lambda \circ \lambda S$: let $l = [f_1, \dots, f_n] \in TSSX$. We have:

$$\begin{aligned}
(\lambda \circ T\mu_S)(l) &= \lambda(T\mu_S(l)) \\
&= \lambda([\mu_S(f_i) \mid i = 1, \dots, n]) \\
&= \lambda x : TX. \begin{cases} \prod_{i=1}^n \mu_S(f_i)(x_i) & x = [x_1, \dots, x_n] \\ 0 & \text{otherwise} \end{cases} \\
&= \lambda x : TX. \begin{cases} \prod_{i=1}^n \sum_{g \in SX} f_i(g) \cdot g(x_i) & x = [x_1, \dots, x_n] \\ 0 & \text{otherwise} \end{cases} \\
&= \lambda x : TX. \begin{cases} \sum_{g_1, \dots, g_n \in SX} \prod_{i=1}^n f_i(g_i) \cdot g_i(x_i) & x = [x_1, \dots, x_m] \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

$$\begin{aligned}
(\mu_S T \circ S\lambda \circ \lambda S)(l) &= \mu_S T(S\lambda(\lambda S(l))) \\
&= \mu_S T \left(S\lambda \left(\lambda y : T S X. \begin{cases} \prod_{i=1}^n \sum_{g \in S X} f_i(g) \cdot g(y_i) & y = [y_1, \dots, y_n] \\ 0 & \text{otherwise} \end{cases} \right) \right) \\
&= \mu_S T \left(\lambda h : S T X. \sum_{g_1, \dots, g_n \in S X \wedge \lambda([g_1, \dots, g_n]) = h} \prod_{i=1}^n f_i(g_i) \cdot g_i(x_i) \right) \\
&= \lambda x : T X. \sum_{g_1, \dots, g_n \in S X} \left(\prod_{i=1}^n f_i(g_i) \right) \cdot \lambda([g_1, \dots, g_n])(x) \\
&= \lambda x : T X. \begin{cases} \sum_{g_1, \dots, g_n \in S X} \left(\prod_{i=1}^n f_i(g_i) \right) \cdot \left(\prod_{i=1}^n g_i(x_i) \right) & x = [x_1, \dots, x_n] \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

Therefore, we have $(\lambda \circ T\mu_S)(l) = (\mu_S T \circ S\lambda \circ \lambda S)(l)$.

- $\lambda \circ \mu_T S = S\mu_T \circ \lambda T \circ T\lambda$: let $l = [[f_{11}, \dots, f_{1m_1}], \dots, [f_{n1}, \dots, f_{nm_n}]] \in T T S X$. We have:

$$\begin{aligned}
(\lambda \circ \mu_T S)(l) &= \lambda(\mu_T S(l)) \\
&= \lambda(\#_{i=1}^n [f_{i1}, \dots, f_{im_i}]) \\
&= \lambda x : T X. \begin{cases} \prod_{i=1}^n \prod_{j=1}^{m_i} f_{ij}(x_{ij}) & x = \#_{i=1}^n [x_{i1}, \dots, x_{im_i}] \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

$$\begin{aligned}
(S\mu_T \circ \lambda T \circ T\lambda)(l) &= S\mu_T(\lambda T(T\lambda(l))) \\
&= S\mu_T(\lambda T([\lambda([f_{11}, \dots, f_{1m_1}]), \dots, \lambda([f_{n1}, \dots, f_{nm_n}])])) \\
&= S\mu_T \left(\lambda T \left(\left[\lambda y : T X. \begin{cases} \prod_{j=1}^{m_1} f_{1j}(y_j) & y = [y_1, \dots, y_{m_1}] \\ 0 & \text{otherwise} \end{cases}, \dots, \lambda y : T X. \begin{cases} \prod_{j=1}^{m_n} f_{nj}(y_j) & y = [y_1, \dots, y_{m_n}] \\ 0 & \text{otherwise} \end{cases} \right] \right) \right) \\
&= S\mu_T \left(\lambda h : S T X. \sum_{i=1}^n \sum_{y_1, \dots, y_{m_i} \in T X} \begin{cases} \prod_{j=1}^{m_i} f_{ij}(y_j) & h = \lambda y : T X. \begin{cases} \prod_{j=1}^{m_i} f_{ij}(y_j) & y = [y_1, \dots, y_{m_i}] \\ 0 & \text{otherwise} \end{cases} \\ 0 & \text{otherwise} \end{cases} \right) \\
&= \lambda x : T X. \begin{cases} \sum_{i=1}^n \sum_{y_1, \dots, y_{m_i} \in T X} \begin{cases} \prod_{j=1}^{m_i} f_{ij}(y_j) & y_i = [y_{i1}, \dots, y_{im_i}] \\ 0 & \text{otherwise} \end{cases} & x = \#_{i=1}^n [y_1, \dots, y_{m_i}] \\ 0 & \text{otherwise} \end{cases} \\
&= \lambda x : T X. \begin{cases} \prod_{i=1}^n \prod_{j=1}^{m_i} f_{ij}(x_{ij}) & x = \#_{i=1}^n [x_{i1}, \dots, x_{im_i}] \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

Therefore, we have $(\lambda \circ \mu_T S)(l) = (S\mu_T \circ \lambda T \circ T\lambda)(l)$.

□

Corollary 49

$(ST, \eta_{\mathbf{SG}+\mathbf{CSG}}, \mu_{\mathbf{SG}+\mathbf{CSG}})$ is a monad on **Set**, where

$$\mu_{\mathbf{SG}+\mathbf{CSG}} : STST \xrightarrow{S\lambda T} SSTT \xrightarrow{\mu_S \mu_T} ST$$

Another approach to get a monad structure on ST is to use monads in extension form.

[MW10] shows¹ that we can define an **SG**-algebra $(STX, (-)^\lambda)$ where for all $f : X \rightarrow STY$, we have $f^\lambda : TX \rightarrow STY$ defined by $f^\lambda([x_1, \dots, x_n]) = \lambda([f(x_1), \dots, f(x_n)])$. It is then showed that this **SG**-algebra gives a distributive law of the monad T over the monad S , and therefore a monad structure on ST .

In extensive form, the monad is $(ST, \eta_{\mathbf{SG}+\mathbf{CSG}}, (-)^{(ST)})$ where for all $f : X \rightarrow STY$, we have $f^{(ST)} : STX \rightarrow STY$ defined by $f^{(ST)} = (f^\lambda)^T$.

We also have that $\lambda = (S\eta_T)^\lambda$ as expected.

2.3 Monoids

Now that we have the free semigroup, we can easily construct the free monoid by adding a new constant (arity 0 operator) to the signature, and adding the corresponding axioms.

Free commutative monoids

Let X be a set. We consider the theory of commutative monoids **CM**, with a signature Σ with two operators: $(+)$ of arity 2, and 0 of arity 0, and with the three axioms

$$\begin{aligned} a, b &\vdash a + b = b + a \\ a, b, c &\vdash a + (b + c) = (a + b) + c \\ a &\vdash a + 0 = a = 0 + a \end{aligned}$$

We pose²

$$SX = \{f : X \rightarrow \mathbb{N} \mid f \text{ has finite support}\}$$

There is no non-empty support condition as in the free commutative semigroup, which means that the function f can be the constant 0 function. This corresponds to the new constant 0 in the signature.

We pose

$$\mathbf{0} = \lambda x. 0 \in SX$$

.

We keep the same definition for the operator $+$ and for $\eta_{\mathbf{CM}}$ as in the free commutative semigroup.

We pose $F_{\mathbf{CM}}X = (SX, (+), \mathbf{0})$ where $+$ and $\eta_{\mathbf{CM}}$ are defined as above.

We want to show that $F_{\mathbf{CM}}X$ is a free **CM**-model over X .

- ▷ $F_{\mathbf{CM}}X$ is a **CM**-model: the axioms of **CM** are valid in $F_{\mathbf{CM}}X$ because of the commutativity and associativity of addition in $\mathbb{N}$, and because of the existence of the constant function $\mathbf{0}$ which acts as the identity for $+$.
- ▷ Let $A = (\underline{A}, +_A, 0_A)$ be a **CM**-model over X with environment $\varphi_A : X \rightarrow U A$. We define $\Sigma\varphi_A : SX \rightarrow \underline{A}$ for all $f \in SX$ by

$$\Sigma\varphi_A(f) = \sum_{x \in X} f \ x \cdot \varphi_A(x)$$

We have $\Sigma\varphi_A(\mathbf{0}) = 0_A$.

For the sake of doing the proof at least once, we'll prove that $\Sigma\varphi_A$ is well defined on $SX \setminus \{\mathbf{0}\}$.

¹example 8.5

²we may want to find a different name, as it may be confused with the one for the commutative semigroup

Proposition 50

For all $f \in SX$, the sum $\sum_{x \in X} f \cdot x \cdot \varphi_A(x)$ is well defined.

Proof

Let $f \in SX \setminus \{0\}$. We note $\text{Supp } f = \{x \in X \mid f(x) \neq 0\}$.

Since f has finite support, $\text{Supp } f$ is a finite set. Moreover, since $f \neq 0$, $\text{Supp } f$ is non-empty. Therefore, we can write $\text{Supp } f = \{x_1, x_2, \dots, x_n\}$ for some $n \geq 1$ and we have a bijection σ between $\{1, 2, \dots, n\}$ and $\text{Supp } f$ given by $i \mapsto x_i$.

Therefore, we can rewrite the sum as follows:

$$\begin{aligned} \sum_{x \in X} f \cdot x \cdot \varphi_A(x) &= \sum_{x \in \text{Supp } f} f \cdot x \cdot \varphi_A(x) \\ &= \sum_{i=1}^n f \cdot \sigma(i) \cdot \varphi_A(\sigma(i)) \end{aligned}$$

The definition is valid because $+_A$ is associative.

Let σ' be another bijection between $\{1, 2, \dots, n\}$ and $\text{Supp } f$. Let's show that we get the same value for the sum if we use σ' instead of σ . This is true because σ and σ' are bijections, so there exists a permutation π of $\{1, 2, \dots, n\}$ such that $\sigma' = \sigma \circ \pi$. Therefore, we have:

$$\begin{aligned} \sum_{i=1}^n f \cdot \sigma'(i) \cdot \varphi_A(\sigma'(i)) &= \sum_{i=1}^n f \cdot \sigma(\pi(i)) \cdot \varphi_A(\sigma(\pi(i))) \\ &= \sum_{i=1}^n f \cdot \sigma(i) \cdot \varphi_A(\sigma(i)) \quad \text{by commutativity of } +_A \end{aligned}$$

Therefore, the definition of $\Sigma\varphi_A(f)$ does not depend on the choice of the bijection between $\{1, 2, \dots, n\}$ and $\text{Supp } f$, and is well defined.

Now if $f = 0$, we have $\text{Supp } f = \emptyset$, and the sum is the empty sum, which is defined to be 0_A . $\square$

We can now prove that $\Sigma\varphi_A$ is the unique morphism h of **CM**-models from $F_{\mathbf{CM}}X$ to A such that the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{\eta_{\mathbf{CM}}} & SX \\ & \searrow \varphi_A & \downarrow h \\ & & UA \end{array}$$

– Preservation of the operator $+$: let $f, g \in SX$. We have:

$$\begin{aligned} \Sigma\varphi_A(f + g) &= \Sigma\varphi_A(\lambda x. f \cdot x + g \cdot x) \\ &= \sum_{x \in X} (f \cdot x + g \cdot x) \cdot \varphi_A(x) && \text{by definition} \\ &= \sum_{x \in X} f \cdot x \cdot \varphi_A(x) + \sum_{x \in X} g \cdot x \cdot \varphi_A(x) && \text{since both } f \text{ and } g \text{ have finite support} \\ &= \Sigma\varphi_A(f) + \Sigma\varphi_A(g) \end{aligned}$$

- Commutation of the diagram: $\eta_{\mathbf{CM}} = \eta_{\mathbf{CSG}}$, so the proof is the same as for the free commutative semigroup.
- Uniqueness: let $h : F_{\mathbf{CM}}X \rightarrow A$ be a morphism of **CM**-models such that the diagram above commutes. We want to show that $h = \Sigma\varphi_A$. Let $f \in SX$. We'll show this lemma first:

Lemma 51

For all $f \in SX$, we have $f = \sum_{x \in X} f \ x \cdot \eta_{\mathbf{CM}}(x)$.

Proof

Let $f \in SX$. We have $\text{Supp } f = \{x \in X \mid f(x) \neq 0\}$. Since f has finite support, $\text{Supp } f$ is a finite set. Moreover, since f is not necessarily non-zero, $\text{Supp } f$ can be empty.

If $\text{Supp } f$ is empty, then $f = \mathbf{0}$ and we have:

$$\sum_{x \in X} f \ x \cdot \eta_{\mathbf{CM}}(x) = \sum_{x \in X} 0 \cdot \eta_{\mathbf{CM}}(x) = 0 = f$$

If $\text{Supp } f$ is non-empty, we can write $\text{Supp } f = \{x_1, x_2, \dots, x_n\}$ for some $n \geq 1$. We'll prove by induction on n that $f = \sum_{x \in X} f \ x \cdot \eta_{\mathbf{CM}}(x)$.

- * If $n = 1$, then $\text{Supp } f = \{x_1\}$ and we have $f = f \ x_1 \cdot \eta_{\mathbf{CM}}(x_1) = \sum_{x \in X} f \ x \cdot \eta_{\mathbf{CM}}(x)$.
- * If $n > 1$, we have $\text{Supp } f = \{x_1, x_2, \dots, x_n\}$.

We can write $f = f \ x_1 \cdot \eta_{\mathbf{CM}}(x_1) + f'$ where $f' = \lambda x. \begin{cases} 0 & \text{if } x = x_1 \\ f \ x & \text{otherwise} \end{cases}$

We have $\text{Supp } f' = \{x_2, \dots, x_n\}$, so by the induction hypothesis, we have $f' = \sum_{x \in X} f' \ x \cdot \eta_{\mathbf{CM}}(x)$. Therefore, we have:

$$\begin{aligned} f &= f \ x_1 \cdot \eta_{\mathbf{CM}}(x_1) + f' \\ &= f \ x_1 \cdot \eta_{\mathbf{CM}}(x_1) + \sum_{x \in X} f' \ x \cdot \eta_{\mathbf{CM}}(x) \\ &= f \ x_1 \cdot \eta_{\mathbf{CM}}(x_1) + \sum_{x \in X} f \ x \cdot \eta_{\mathbf{CM}}(x) - f \ x_1 \cdot \eta_{\mathbf{CM}}(x_1) \quad \text{by definition of } f' \\ &= \sum_{x \in X} f \ x \cdot \eta_{\mathbf{CM}}(x) \end{aligned}$$

□

We then have:

$$\begin{aligned} h(f) &= h\left(\sum_{x \in X} f \ x \cdot \eta_{\mathbf{CM}}(x)\right) \\ &= \sum_{x \in X} f \ x \cdot h(\eta_{\mathbf{CM}}(x)) \\ &= \sum_{x \in X} f \ x \cdot \varphi_A(x) && \text{since the diagram commutes} \\ &= \Sigma \varphi_A(f) \end{aligned}$$

Therefore we proved that $F_{\mathbf{CM}}X$ is a free $\mathbf{CM}$ -model over X .

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- [MW10] Francisco Marmolejo and R. J. Wood. “Monads as Extension Systems—No Iteration is Necessary”. In: *Theory and Applications of Categories* (2010). URL: <https://api.semanticscholar.org/CorpusID:18849266>.

A DAILY RECAP

- 05/05/2026: first day, talk with Ohad, building tour, admin stuff, setting up the git repo and starting to read the notes of Ohad
- 06/05/2026: reading Ohad’s notes, starting to write my own notes. lots of definitions to try and understand how things work.
- 07/05/2026: read the paper on frexes and frals, wrote down the relevant definitions. Meeting with Ohad: 3 equivalent definitions of adjunctions $\Rightarrow$ universal arrows, examples where/when they are used + equivalence of definitions of free algebras and how to construct a free algebra for (commutative) semigroups.
 Gave examples of uses of universal arrows: defining adjunctions, defining products and coproducts, and defining free algebras.
 Defined and proved the free algebra for commutative semigroups, started to work on the free algebra for semigroups.
- 08/05/2026: Defined and proved the free algebra for semigroups and commutative monoids. Proved the details that were missing for the free algebra for commutative semirings (in the commutative monoid part).
 Started to work on the free algebra for the combination of semigroups and commutative semigroups, but still need to define the product operator and check that the distributivity axioms hold, and that the resulting algebra is indeed a free algebra for the combination.
- 11/05/2026: finished the definition of the product operator for the free algebra for the combination of semigroups and commutative semigroups, and proved that it is indeed a free algebra for the combination.
 Defined multiplication for the free semigroup and commutative semigroup, and proved that we get a monad structure on **Set** in both cases.
 Defined the distributive law of the monad for the free semigroup over the monad for the free commutative semigroup, and proved that it is indeed a distributive law $\rightarrow$ we get a monad structure on **Set** for the free algebra for the combination of semigroups and commutative semigroups.
- 12/05/2026: definition of Kleisli category and related objects, research and definition of algebra and distributive laws of monads in extension form, and link with the previous definition of distributive laws. Example 8.5 of [MW10] is almost the same thing as Section 2.2.

B TODO

- follow Ohad’s abstract picture

- dig in the paper about frexes to learn more about that (both frexes and frals) and write the relevant definitions down
- write the definition of free extensions with universal arrows
- look at what was done with involution to take inspiration

C QUESTIONS FOR OHAD

- what exactly is meant by an operad spanned by an operation (e.g what is the operad $\mathcal{O}$ spanned by $(\cdot) : 2$)